

Week 13 – Backend Security & Isolation

Summary:

Week 13 focused on **improving backend security** and isolating program execution to prevent system-level issues.

Key Activities:

- Implemented **sandboxing mechanisms** inside Docker containers.
- Verified that each program execution is **isolated** to prevent interference.
- Checked file permissions and restricted container access to sensitive directories.
- Monitored for security loopholes during multi-language execution.

Technologies Used: Python, FastAPI, Docker, WSL2, Linux Permissions

Outcome:

Backend is now **secure**, isolated, and ready to safely execute programs from multiple users simultaneously.